Beyond the 4Cs: An Analysis of Provenance, Production Method, and the Distinct Structure of Diamond Pricing

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Introduction and Objectives

For many centuries, diamonds have occupied a peculiar economic position as a commodity whose price is shaped as much by narrative, provenance, and perceived rarity as by their measurable, physical properties. In recent years, that has become harder to justify, as lab-grown diamonds, chemically and visually indistinguishable from their mined counterparts, yet produced in bulk and on demand rather than extracted from the earth, have entered the market at a fraction of the price of natural stones. For diamond-producing economies such as Botswana, whose entire economic infrastructure was built on the prestige of natural diamonds, the rise of lab-grown diamonds has created a serious economic problem. Diamonds account for roughly 80% of Botswana's export earnings and around a quarter to a third of its GDP (World Bank, 2025), meaning that any erosion of the natural diamond's premium is not just a jewelry industry concern, but a national economic one.

As a citizen of Botswana, this project's motivation stems from my interest in the diamond industry and curiosity regarding the pricing structures of diamonds. For years, we have been told that the price of a diamond is determined purely by the 4Cs: carat, cut, clarity, and color. But with the introduction of lab-grown diamonds in recent years, their significantly lower prices raise questions about the honesty of the pricing narrative we were sold, and what exactly the disparities in pricing between natural and lab-grown options really are. This project tests what factors actually drive a diamond's price, and whether geographic origin and production method

survive that test as meaningful factors once the diamond's physical characteristics are accounted for.

The first question this project asks is whether Botswana's status as the world's second largest natural diamond producer has a significant and measurable effect on the price of the stone once carat weight and quality are controlled for, or whether it prices about the same as other natural origins like Angola and Canada. The second question is whether natural and lab-grown diamonds are priced using the same underlying logic, or whether the two markets value different characteristics entirely.

Data

All data for this project comes from Brilliant Earth, a United States based online diamond retailer, and was collected using Python through their publicly accessible product API. The dataset contains 187,579 diamond listings across 26 variables, split between 94,448 natural diamonds and 93,131 lab-grown diamonds. Natural diamonds were filtered by geographic origin into four groups: Botswana, Angola, Canada, and Recycled. Recycled diamonds made up only 143 of the natural stones in the dataset and are noted throughout this project as an underpowered group whose results should be interpreted with caution.

Each listing includes the diamond's price, carat weight, cut (Fair; worst → Super Ideal; best), color (K; worst/most tinted → D; best/colorless), clarity (SI2; worst/heavily flawed → FL; best/flawless), and shape (Round, Oval, Princess and seven others), along with additional quality measures such as polish, symmetry, and fluorescence. It is worth noting that color and clarity grades in this dataset are made of a subset of the full GIA scale, reflecting Brilliant Earth's retail quality floor. Physical measurements including depth, table percentage (the width of its flat top divided by its total average diameter), length, and length to width ratio are also recorded, as well

as the certifying report used to grade the stone (GIA, IGI, HRD, or GCAL). A number of variables were engineered from these raw fields to support the analysis, including a log transformed price and carat (log_price, log_carat) to correct for their right-skewed distributions and stabilize variance, numeric ordinal codes for cut, color, and clarity (cut_n, color_n, clarity_n), price per carat, and an origin group variable that combines diamond type and country of origin into a single variable within the category: Angola, Botswana, Canada, Recycled, and Lab Grown.

Two data quality issues were identified and corrected prior to the analysis. The first issue was that the polish and symmetry variables each contained a small number of rows using inconsistent category labels, such as "Excl" instead of "Excellent" and "Vgood" instead of "Very Good." These were merged into their correct categories. Secondly, the polish variable's "Fair" category contained only 3 of the 187,579 total rows, too few to estimate a stable effect in a regression model, so it was combined with "Good" into a single "Good or Below" category. The symmetry variable's "Fair" category, by contrast, contained 55 rows and was kept separate, since it represented a real, albeit small, subgroup rather than a data entry issue.

The fluorescence variable was excluded from all formal models due to a significant amount of missing values, with roughly 85% of rows missing. The certifier/report variable (GIA, IGI, HRD, GCAL) was also excluded from all relevant models involving diamond type or origin, since it is closely tied to those variables. The majority of lab-grown diamonds in this dataset were certified by IGI, while majority of natural diamonds were certified by GIA, meaning report and diamond type carry a lot of the same information.

Analysis and Methodology

To test the first objective, price differences by diamond type and by origin were first examined using one-way ANOVA in JMP, before moving to a controlled multiple regression model. This was an important first step because when each variable is run on their own, type and origin appear to barely make much difference to the price, and it is only once carat weight and quality are accounted for that their real effect becomes visible. See Appendix A for a comparison table of all models discussed, with full JMP output for each provided in Appendix B.

A one-way ANOVA of price by diamond type showed that natural diamonds had an average price of $5,081.62 compared to $3,880.03 for lab-grown diamonds, a difference of $1,201.59 ($F = 638.33$, $p < .0001$). Levene's test showed that the variance in price differs significantly between the two groups ($p < .0001$), so a Welch's test was also run to confirm the result without assuming equal variances ($F = 645.57$, $p < .0001$). Despite this large and highly significant difference in means, diamond type alone explained almost none of the variation in price ($R^2 = 0.0034$). A variable this significant explaining this little of the variance is a sign that something else, namely carat weight, is confounding the relationship and needs to be controlled for.

The same pattern held when comparing origin among natural diamonds only. A one-way ANOVA of price by origin group (Angola, Botswana, Canada, and Recycled) was also highly significant ($F = 40.23$, $p < .0001$) but again explained almost none of the variation in price on its own ($R^2 = 0.0013$). A Tukey HSD test showed that Angola ($5,587.51), Botswana ($4,946.54), and Canada ($3,967.39) were all significantly different from one another, with Angola pricing highest among the three. Recycled diamonds had the highest average price ($6,854.65) but could not be statistically distinguished from any of the other three groups, since the group contains

only 143 of the 94,448 natural diamonds in the dataset and carries a wide confidence interval as a result. Recycled diamonds are treated as an underpowered group throughout this project for this reason.

Together, these two ANOVA test results tell a surface level story: type and origin are statistically real but alone, they appear to explain almost nothing about the price of a diamond. A multiple linear regression model was used next to test whether that story changes once carat weight and diamond quality are held constant, and to begin to quantify that change.

The regression model used log_carat, origin_group, and the diamond's quality characteristics (cut, color, clarity, polish, and symmetry, each represented by their numeric ordinal codes) to predict log_price. Diamond type was not included directly, since origin_group already separates lab-grown diamonds from each natural origin within a single variable, and including both would have caused the model to fail due to perfect overlap between the two.

This baseline model explained most of the price variation ($R^2 = 0.9001$), a dramatic increase from the ANOVA results. Carat weight, origin, color, and clarity were all highly significant ($p < .0001$), while cut was not a significant predictor of price once the other variables were accounted for ($p = 0.2088$). Origin group also remained highly significant after controlling for quality.

Initial parameter estimates for polish and symmetry showed severe multicollinearity, with variance inflation factors in the hundreds. This traced to a structural pattern rather than a data error: symmetry and cut grades differ systematically by origin, with lab-grown diamonds graded "Excellent" symmetry in 89.5% of cases compared to 54 to 67% for natural origins, reflecting the greater consistency of lab-grown production. A sensitivity check excluding Recycled diamonds confirmed this collinearity was structural, not driven by that small subgroup.

To address this without discarding real information, Principal Components Analysis was run on log_carat, cut_n, color_n, clarity_n, and numeric ordinal versions of polish and symmetry. The first component explained 36% of the combined variance and loaded heavily on cut, polish, and symmetry (0.67 to 0.84), confirming these three represent one underlying cutting-precision variable, which replaced them in the final model. Carat, color, and clarity were retained individually, and shape was added separately as a predictor. The resulting model achieved an R^2 of 0.9023, comparable to the baseline's 0.9001, confirming the fix preserved explanatory power while resolving the multicollinearity.

The final model, incorporating the principal component and diamond shape, explained the majority of the variation in price ($R^2 = 0.9023$, $p < .0001$). All five main predictors remained highly significant. Holding carat, color, clarity, cutting precision, and shape constant, origin_group's coefficients showed Angola with the largest premium over the reference group (0.2589), followed closely by Botswana (0.2207) and Canada (0.2156), while lab-grown diamonds carried a substantial discount (-1.1291). This confirms the pattern first seen in the ANOVA: Botswana prices below Angola even after controlling for quality, and the gap between any natural origin and lab-grown dwarfs the differences among natural origins themselves. Carat remained the single strongest predictor (coefficient 1.5604), consistent with its dominance throughout this analysis. A supporting logistic regression, using only physical and quality characteristics to predict whether a natural diamond was Botswana-origin, achieved modest discrimination (AUC = 0.731, 67.7% accuracy), suggesting Botswana diamonds are not strongly distinguishable from other natural origins on physical grounds alone.

To test the second objective, stepwise regression was used to let JMP select the strongest predictors of log_price automatically, first on the full dataset and then two separate ones on

natural and lab-grown diamonds respectively. Comparing which variables each model selected, and in what order, revealed whether the same characteristics drove the price in both markets, or whether each market prices diamonds in completely different ways.

On the full dataset, carat entered first and alone explained 63% of price variation, followed immediately by the natural-versus-lab-grown split, which added another 24 percentage points, more than every quality and shape variable combined. Origin, color, clarity, shape, symmetry, and polish all entered afterward with progressively smaller contributions, and the final model reached an R^2 of 0.9025.

Splitting the data by diamond type revealed real differences in how each market priced a stone. For natural diamonds, color entered before clarity, and cut entered fourth, immediately after the top three predictors (R^2 Adj = 0.9481). For lab-grown diamonds, clarity entered before color, and cut did not enter until much later in the selection process, contributing almost nothing to the model (R^2 Adj = 0.9439). Diamond shape also carried far more weight in the natural model than the lab-grown one, consistent with shape depending partly on the rough stone's original form in natural diamonds, while lab-grown stones are grown directly to a target shape.

These differences answered the second objective directly: natural and lab-grown diamonds were not priced using the same logic. Carat dominated both markets, but beyond that, natural diamonds were valued more by color and cutting quality, while lab-grown diamonds were valued more by clarity, with cut playing almost no role. This supports treating natural and lab-grown diamonds as related but distinct products, rather than direct substitutes priced on the same scale.

Shortcomings and Limitations

The main limitation is in the source of the data: it comes exclusively from one retailer, Brilliant Earth, rather than a random and representative sample of the broader diamond market. A representative sample would draw diamonds from across multiple retailers, price points, and sourcing channels; this dataset instead reflects one company's existing inventory, already shaped by its own business decisions about which quality tiers to stock and which countries to source from. Prices, quality distributions, and even the range of color and clarity grades supplied may not apply to other retailers, auction markets, or private sales, particularly given this retailer's stock, as indicated in the Data section, only spans a fraction of the entire grading scale. For this reason, any conclusions about origin premiums or market pricing logic should be understood as specific to this retail context, not the entire diamond market.

Beyond the data source itself, the size of the dataset introduces its own caveat: with 187,579 observations, even very small effects reach statistical significance, which is why this project prioritized effect size and R^2 contribution over p-values alone when judging whether a variable actually mattered or not. Residual plots across the regression models also showed some spread that widened at lower predicted values, a mild violation of the constant-variance assumption. Neither issue biases the coefficient estimates themselves, but together they mean results should be read as strong evidence of direction and relative importance rather than exact probability statements, alongside the underpowered Recycled origin group and origin_group's elevated variance inflation factors already addressed in the Analysis and Methodology section.

Discussion

This project began by questioning whether a diamond's price is really explained by the traditional 4Cs. The results suggest it is not, but not for the reason one might expect. Carat

weight remained the dominant driver of price in every model, and physical quality clearly mattered. But once carat and quality were held constant, the diamond market split cleanly in two. Natural and lab-grown diamonds are not one product sold at two price points; they are two markets valuing different things. Natural diamond pricing leaned more on color and cutting precision, both shaped by what a rough stone allowed a cutter to do with it. Lab-grown pricing leaned on clarity and barely responded to cut at all, consistent with a product built to specification rather than shaped by nature. The diamond market is not one market that happens to include two production methods. It is two markets that happen to sell a visually similar product.

For Botswana, this is both encouraging and cautionary. Origin survived statistical control, meaning a natural diamond's provenance still carries real, measurable price value beyond what a gemologist can grade, and that value is not something lab-grown diamonds can claim. But Botswana did not command the largest premium among natural origins in this dataset: Angola priced consistently higher, and Canada's premium was nearly identical to Botswana's own. The story protecting Botswana's diamond economy is not a Botswana story specifically. It is a natural diamond story that Botswana shares with every other natural origin here, and one that shrinks every time a lab-grown stone takes a sale that would have gone to a mined one.

That distinction should shape how Botswana positions itself. A strategy built on "this diamond is from Botswana" competes on geography, and this analysis shows that story alone does not command a uniquely large premium. A strategy built on the individuality of the natural stone itself, the fact that no two rough diamonds are alike in a way no factory can replicate, asks consumers to value something lab-grown diamonds are structurally unable to offer. For an economy where diamonds account for roughly a fifth to a third of GDP, that is the difference

between defending a premium this analysis shows is thin, and defending one this analysis shows is real.

As lab production continues to improve and the physical differences between natural and lab-grown diamonds become harder to detect, the line between the two markets may depend less on measurable quality and more on consumer perception. Whether that perception remains strong enough to preserve the natural diamond's premium is an open question, and for Botswana, the answer to it extends well beyond the jewelry industry.

APPENDICES

Appendix A – Model Results

Model	Response	Key Result	R^2 / Fit Statistic	Note
ANOVA — Type	price	Natural vs. Lab Grown means differ ($5,081.62 vs. $3,880.03)	R^2 = 0.0034	Confounded surface story
ANOVA — Origin	price	Angola > Botswana > Canada (Natural only)	R^2 = 0.0013	Recycled underpowered (n=143)
Final MLR	log_price	carat, origin, color, clarity, Prin1 (cutting precision), shape	R^2 = 0.9023	Reported model, Objective 1
Stepwise — Full Dataset	log_price	Automated selection; type/origin split adds 24 pts after carat	R^2 = 0.9025	Origin enters 2nd
Stepwise — Natural Only	log_price	Color enters before clarity; cut enters 4th	R^2 Adj = 0.9481	Objective 2
Stepwise — Lab-Grown Only	log_price	Clarity enters before color; cut enters late, minimal effect	R^2 Adj = 0.9439	Objective 2
Logistic Regression	Botswana_Dummy	Physical/quality traits + shape predict Botswana origin	AUC = 0.731, Accuracy = 67.7%	Supports Objective 1

Appendix B – Full JMP Output

B.1 - ANOVA: Price by Diamond Type

Oneway Analysis of price By diamond_type

Oneway Anova

Summary of Fit

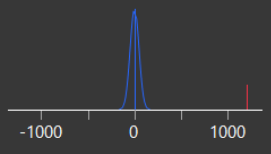
Rsquare	0.003391
Adj Rsquare	0.003386
Root Mean Square Error	10298.77
Mean of Response	4485.047
Observations (or Sum Wgts)	187579

Pooled t Test

Natural-Lab Grown

Assuming equal variances

Difference	1201.59	t Ratio	25.26517
Std Err Dif	47.56	DF	187577
Upper CL Dif	1294.80	Prob > |t|	<.0001*
Lower CL Dif	1108.37	Prob > t	<.0001*
Confidence	0.95	Prob < t	1.0000



Cohen's d 0.117

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio	Prob > F
diamond_type	1	6.7704e+10	6.77e+10	638.3287	<.0001*
Error	187577	1.9895e+13	106064698		
C. Total	187578	1.9963e+13			

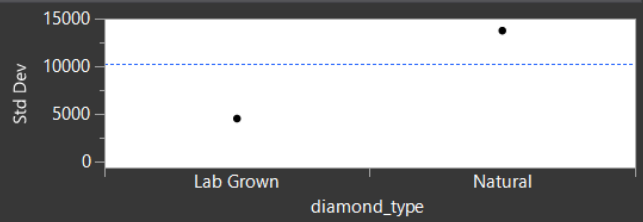
Means for Oneway Anova

Level	Number	Mean	Std Error	Lower 95%	Upper 95%
Lab Grown	93131	3880.03	33.747	3813.9	3946.2
Natural	94448	5081.62	33.511	5015.9	5147.3

Std Error uses a pooled estimate of error variance

Oneway Analysis of price By diamond_type

Tests that the Variances are Equal



Level	Count	Std Dev	MeanAbsDif to Mean	MeanAbsDif to Median
Lab Grown	93131	4554.31	2956.570	2612.258
Natural	94448	13791.23	5083.909	4028.237

Test	F Ratio	DFNum	DFDen	p-Value
O'Brien[.5]	186.5794	1	187577	<.0001*
Brown-Forsythe	939.0950	1	187577	<.0001*
Levene	2392.2146	1	187577	<.0001*
Bartlett	96819.968	1	.	<.0001*
F Test 2-sided	9.1698	94447	93130	<.0001*

Welch's Test

Welch Anova testing Means Equal, allowing Std Devs Not Equal

F Ratio	DFNum	DFDen	Prob > F
645.5703	1	115066	<.0001*

t Test
25.4081

B.2 - ANOVA: Price by Origin Group (Natural Only)

Oneway Analysis of price By origin_group

Oneway Anova

Summary of Fit

Rsquare	0.001276
Adj Rsquare	0.001245
Root Mean Square Error	13782.64
Mean of Response	5081.624
Observations (or Sum Wgts)	94448

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio	Prob > F
origin_group	3	2.2927e+10	7.6423e+9	40.2307	<.0001*
Error	94444	1.7941e+13	189961187		
C. Total	94447	1.7964e+13			

Means for Oneway Anova

Level	Number	Mean	Std Error	Lower 95%	Upper 95%
Angola	35009	5587.51	73.7	5443.1	5731.9
Botswana Sort	49130	4946.54	62.2	4824.7	5068.4
Canada	10166	3967.39	136.7	3699.5	4235.3
Recycled	143	6854.65	1152.6	4595.6	9113.7

Std Error uses a pooled estimate of error variance

Connecting Letters Report

Level				Mean	Std Error
Recycled	A	B	C	6854.7	1152.6
Angola	A			5587.5	73.7
Botswana Sort		B		4946.5	62.2
Canada			C	3967.4	136.7

Levels not connected by same letter are significantly different.

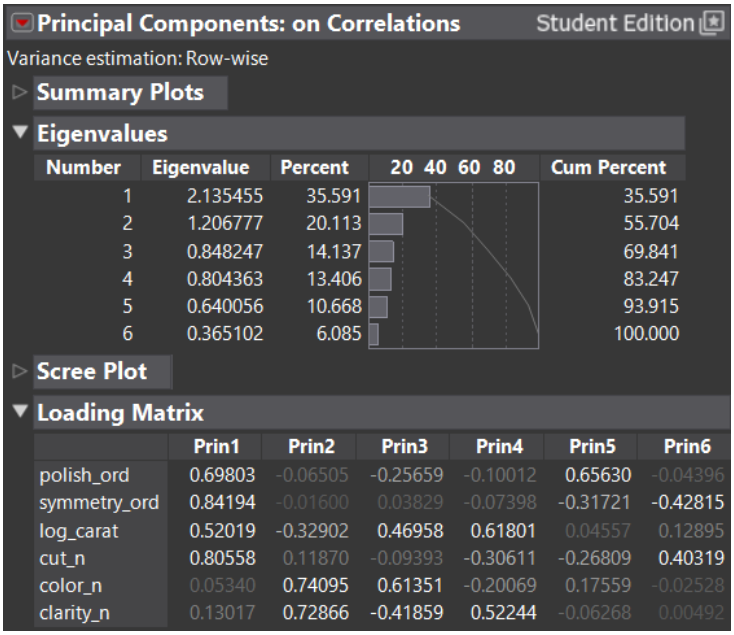
B.3 - Correlation Matrix

Multivariate Student Edition

Correlations

	polish_ord	symmetry_ord	log_carat	cut_n	color_n	clarity_n
polish_ord	1.0000	0.3970	0.2264	0.4157	-0.0319	0.0572
symmetry_ord	0.3970	1.0000	0.3458	0.6078	0.0266	0.0610
log_carat	0.2264	0.3458	1.0000	0.1865	-0.0472	-0.0479
cut_n	0.4157	0.6078	0.1865	1.0000	0.0775	0.0895
color_n	-0.0319	0.0266	-0.0472	0.0775	1.0000	0.1741
clarity_n	0.0572	0.0610	-0.0479	0.0895	0.1741	1.0000

B.4 - Principal Components Analysis



Principal Components: on Correlations — Student Edition

Variance estimation: Row-wise

▷ Summary Plots

▼ Eigenvalues

Number	Eigenvalue	Percent	20 40 60 80	Cum Percent
1	2.135455	35.591		35.591
2	1.206777	20.113		55.704
3	0.848247	14.137		69.841
4	0.804363	13.406		83.247
5	0.640056	10.668		93.915
6	0.365102	6.085		100.000

▷ Scree Plot

▼ Loading Matrix

	Prin1	Prin2	Prin3	Prin4	Prin5	Prin6
polish_ord	0.69803	-0.06505	-0.25659	-0.10012	0.65630	-0.04396
symmetry_ord	0.84194	-0.01600	0.03829	-0.07398	-0.31721	-0.42815
log_carat	0.52019	-0.32902	0.46958	0.61801	0.04557	0.12895
cut_n	0.80558	0.11870	-0.09393	-0.30611	-0.26809	0.40319
color_n	0.05340	0.74095	0.61351	-0.20069	0.17559	-0.02528
clarity_n	0.13017	0.72866	-0.41859	0.52244	-0.06268	0.00492

B.5 - Final Regression Model

Response log_price

Effect Summary

Summary of Fit

RSquare	0.902345
RSquare Adj	0.902337
Root Mean Square Error	0.327869
Mean of Response	7.75568
Observations (or Sum Wgts)	187579

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	17	186304.35	10959.1	101946.9
Error	187561	20162.42	0.107498	Prob > F
C. Total	187578	206466.78		<.0001*

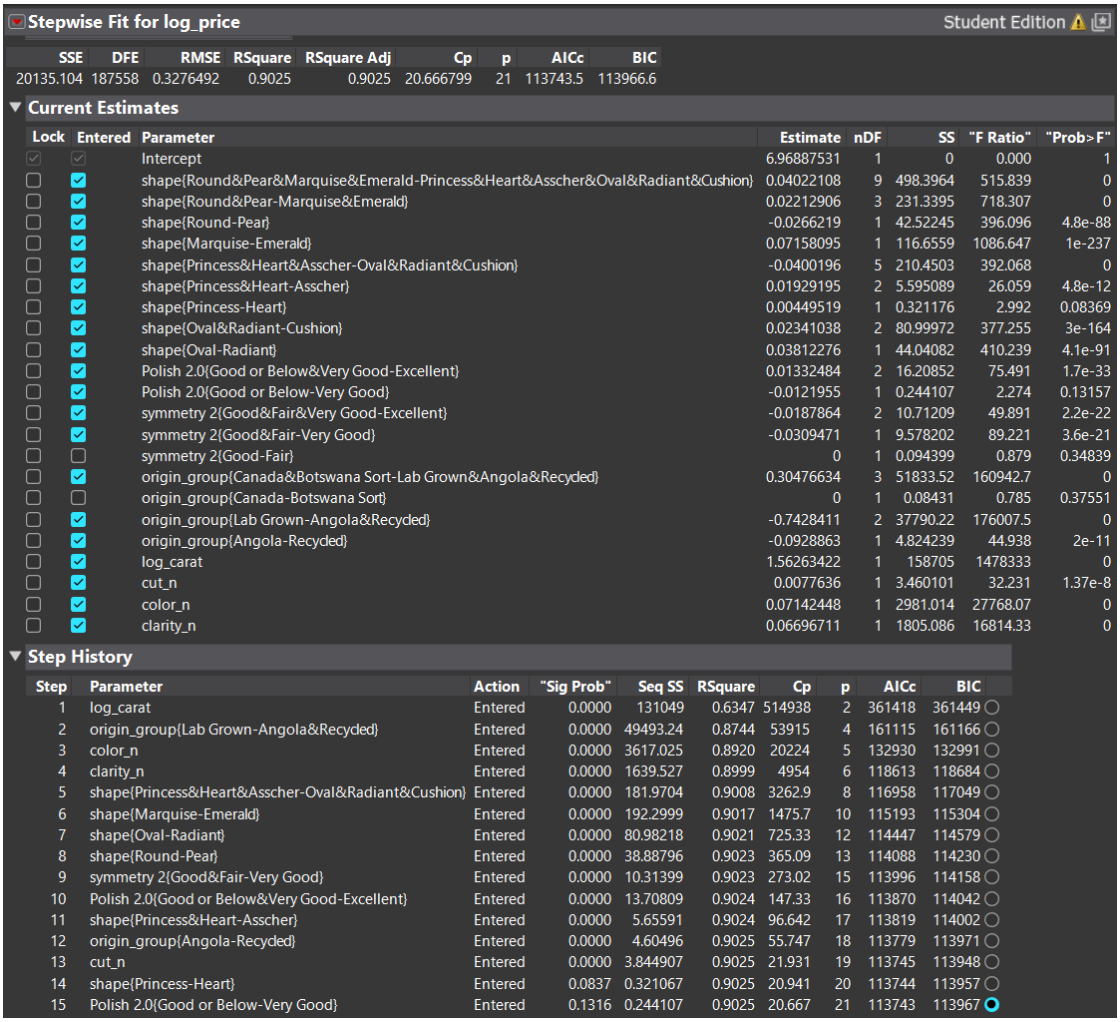
Parameter Estimates

Term	Estimate	Std Error	t Ratio	Prob>|t|	VIF
Intercept	7.0871134	0.006296	1125.6	<.0001*	.
origin_group[Angola]	0.2588717	0.005721	45.25	<.0001*	8.7297224
origin_group[Botswana Sort]	0.2207357	0.005696	38.75	<.0001*	11.009566
origin_group[Canada]	0.2155749	0.00613	35.17	<.0001*	3.4162385
origin_group[Lab Grown]	-1.129052	0.0057	-198.1	<.0001*	14.260662
log_carat	1.5604205	0.001331	1172.6	<.0001*	1.8942643
color_n	0.0717103	0.00043	166.61	<.0001*	1.1766949
clarity_n	0.0668375	0.000524	127.58	<.0001*	1.112245
shape[Asscher]	-0.096749	0.004529	-21.36	<.0001*	9.6360638
shape[Cushion]	-0.019797	0.002796	-7.08	<.0001*	4.6502133
shape[Emerald]	-0.048976	0.002303	-21.27	<.0001*	3.7131824
shape[Heart]	-0.061853	0.003558	-17.38	<.0001*	6.536446
shape[Marquise]	0.0925615	0.003412	27.13	<.0001*	6.1364452
shape[Oval]	0.0656296	0.002064	31.80	<.0001*	3.361182
shape[Pear]	0.0917648	0.002266	40.49	<.0001*	3.684639
shape[Princess]	-0.052254	0.003252	-16.07	<.0001*	5.7157524
shape[Radiant]	-0.010512	0.003005	-3.50	0.0005*	5.133128
Cutting Precision	0.0021279	0.000681	3.12	0.0018*	1.4181437

Effect Tests

Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
origin_group	4	4	53588.83	124627.6	<.0001*
log_carat	1	1	147811.69	1375019	<.0001*
color_n	1	1	2984.06	27759.25	<.0001*
clarity_n	1	1	1749.83	16277.80	<.0001*
shape	9	9	500.15	516.9598	<.0001*
Cutting Precision	1	1	1.05	9.7576	0.0018*

B.6 – Stepwise Regression Model: Full Dataset



Stepwise Fit for log_price — Student Edition

SSE	DFE	RMSE	RSquare	RSquare Adj	Cp	p	AICc	BIC
20135.104	187558	0.3276492	0.9025	0.9025	20.666799	21	113743.5	113966.6

Current Estimates

Lock	Entered	Parameter	Estimate	nDF	SS	"F Ratio"	"Prob>F"
☑	☑	Intercept	6.96887531	1	0	0.000	1
☐	☑	shape{Round&Pear&Marquise&Emerald-Princess&Heart&Asscher&Oval&Radiant&Cushion}	0.04022108	9	498.3964	515.839	0
☐	☑	shape{Round&Pear-Marquise&Emerald}	0.02212906	3	231.3395	718.307	0
☐	☑	shape{Round-Pear}	-0.0266219	1	42.52245	396.096	4.8e-88
☐	☑	shape{Marquise-Emerald}	0.07158095	1	116.6559	1086.647	1e-237
☐	☑	shape{Princess&Heart&Asscher-Oval&Radiant&Cushion}	-0.0400196	5	210.4503	392.068	0
☐	☑	shape{Princess&Heart-Asscher}	0.01929195	2	5.595089	26.059	4.8e-12
☐	☑	shape{Princess-Heart}	0.00449519	1	0.321176	2.992	0.08369
☐	☑	shape{Oval&Radiant-Cushion}	0.02341038	2	80.99972	377.255	3e-164
☐	☑	shape{Oval-Radiant}	0.03812276	1	44.04082	410.239	4.1e-91
☐	☑	Polish 2.0{Good or Below&Very Good-Excellent}	0.01332484	2	16.20852	75.491	1.7e-33
☐	☑	Polish 2.0{Good or Below-Very Good}	-0.0121955	1	0.244107	2.274	0.13157
☐	☑	symmetry 2{Good&Fair&Very Good-Excellent}	-0.0187864	2	10.71209	49.891	2.2e-22
☐	☑	symmetry 2{Good&Fair-Very Good}	-0.0309471	1	9.578202	89.221	3.6e-21
☐	☐	symmetry 2{Good-Fair}	0	1	0.094399	0.879	0.34839
☐	☑	origin_group{Canada&Botswana Sort-Lab Grown&Angola&Recycled}	0.30476634	3	51833.52	160942.7	0
☐	☐	origin_group{Canada-Botswana Sort}	0	1	0.08431	0.785	0.37551
☐	☑	origin_group{Lab Grown-Angola&Recycled}	-0.7428411	2	37790.22	176007.5	0
☐	☑	origin_group{Angola-Recycled}	-0.0928863	1	4.824239	44.938	2e-11
☐	☑	log_carat	1.56263422	1	158705	1478333	0
☐	☑	cut_n	0.0077636	1	3.460101	32.231	1.37e-8
☐	☑	color_n	0.07142448	1	2981.014	27768.07	0
☐	☑	clarity_n	0.06696711	1	1805.086	16814.33	0

Step History

Step	Parameter	Action	"Sig Prob"	Seq SS	RSquare	Cp	p	AICc	BIC
1	log_carat	Entered	0.0000	131049	0.6347	514938	2	361418	361449
2	origin_group{Lab Grown-Angola&Recycled}	Entered	0.0000	49493.24	0.8744	53915	4	161115	161166
3	color_n	Entered	0.0000	3617.025	0.8920	20224	5	132930	132991
4	clarity_n	Entered	0.0000	1639.527	0.8999	4954	6	118613	118684
5	shape{Princess&Heart&Asscher-Oval&Radiant&Cushion}	Entered	0.0000	181.9704	0.9008	3262.9	8	116958	117049
6	shape{Marquise-Emerald}	Entered	0.0000	192.2999	0.9017	1475.7	10	115193	115304
7	shape{Oval-Radiant}	Entered	0.0000	80.98218	0.9021	725.33	12	114447	114579
8	shape{Round-Pear}	Entered	0.0000	38.88796	0.9023	365.09	13	114088	114230
9	symmetry 2{Good&Fair-Very Good}	Entered	0.0000	10.31399	0.9023	273.02	15	113996	114158
10	Polish 2.0{Good or Below&Very Good-Excellent}	Entered	0.0000	13.70809	0.9024	147.33	16	113870	114042
11	shape{Princess&Heart-Asscher}	Entered	0.0000	5.65591	0.9024	96.642	17	113819	114002
12	origin_group{Angola-Recycled}	Entered	0.0000	4.60496	0.9025	55.747	18	113779	113971
13	cut_n	Entered	0.0000	3.844907	0.9025	21.931	19	113745	113948
14	shape{Princess-Heart}	Entered	0.0837	0.321067	0.9025	20.941	20	113744	113957
15	Polish 2.0{Good or Below-Very Good}	Entered	0.1316	0.244107	0.9025	20.667	21	113743	113967

B.7 - Stepwise Regression: Natural Diamonds Only

Stepwise Fit for log_price — Student Edition

Stepwise Regression Control

SSE	DFE	RMSE	RSquare	RSquare Adj	Cp	p	AICc	BIC
6064.1467	94426	0.2534189	0.9481	0.9481	22	22	8756.147	8973.619

Current Estimates

Lock	Entered	Parameter	Estimate	nDF	SS	"F Ratio"	"Prob>F"
☑	☑	Intercept	6.90072346	1	0	0.000	1
☐	☑	shape{Round&Emerald&Pear&Marquise&Princess-Heart&Asscher&Oval&Radiant&Cushion}	0.04431541	9	963.7712	1667.452	0
☐	☑	shape{Round&Emerald-Pear&Marquise&Princess}	0.00774224	4	558.2339	2173.092	0
☐	☑	shape{Round-Emerald}	0.09618893	1	263.2108	4098.506	0
☐	☑	shape{Pear&Marquise-Princess}	0.1745337	2	296.7433	2310.323	0
☐	☑	shape{Pear-Marquise}	-0.0826064	1	74.9872	1167.640	2e-254
☐	☑	shape{Heart&Asscher&Oval&Radiant-Cushion}	0.01966639	4	285.9965	1113.327	0
☐	☑	shape{Heart&Asscher-Oval&Radiant}	-0.0942738	3	230.5007	1196.390	0
☐	☑	shape{Heart-Asscher}	0.05635879	1	17.76529	276.627	5e-62
☐	☑	shape{Oval-Radiant}	0.05190095	1	21.57616	335.967	6.5e-75
☐	☑	Polish 2.0{Good or Below&Very Good-Excellent}	0.06700138	2	44.73524	348.291	2e-151
☐	☑	Polish 2.0{Good or Below-Very Good}	0.05990108	1	4.144332	64.532	9.6e-16
☐	☑	symmetry 2{Good&Fair&Very Good-Excellent}	0.04614572	3	25.39336	131.802	3.3e-85
☐	☑	symmetry 2{Good&Fair-Very Good}	0.06343863	2	14.73972	114.758	1.7e-50
☐	☑	symmetry 2{Good-Fair}	-0.0448274	1	0.38524	5.999	0.01432
☐	☑	origin_group{Canada&Botswana Sort-Angola&Recycled}	-0.0118797	3	37.07397	192.429	2e-124
☐	☑	origin_group{Canada-Botswana Sort}	0.00722849	1	1.587899	24.725	6.62e-7
☐	☑	origin_group{Angola-Recycled}	-0.069028	1	2.582704	40.216	2.3e-10
☐	☑	log_carat	2.0394558	1	95459.62	1486420	0
☐	☑	cut_n	0.07574372	1	183.1484	2851.839	0
☐	☑	color_n	0.09910425	1	3228.682	50274.43	0
☐	☑	clarity_n	0.09309972	1	2275.091	35425.88	0

Step History

Step	Parameter	Action	"Sig Prob"	Seq SS	RSquare	Cp	p	AICc	BIC
1	log_carat	Entered	0.0000	103993.2	0.8892	107332	2	80434.4	80462.8
2	color_n	Entered	0.0000	3665.143	0.9205	50263	3	49036.4	49074.2
3	clarity_n	Entered	0.0000	1980.262	0.9375	19430	4	26404.9	26452.2
4	cut_n	Entered	0.0000	202.9326	0.9392	16272	5	23748.9	23805.6
5	shape{Heart&Asscher-Oval&Radiant}	Entered	0.0000	376.1469	0.9424	10421	8	18621.2	18706.3
6	shape{Round-Emerald}	Entered	0.0000	233.896	0.9444	6783.3	10	15286.2	15390.2
7	shape{Pear&Marquise-Princess}	Entered	0.0000	196.1172	0.9461	3731.5	11	12394.6	12508.1
8	shape{Pear-Marquise}	Entered	0.0000	79.06115	0.9468	2502.4	12	11204.6	11327.5
9	Polish 2.0{Good or Below&Very Good-Excellent}	Entered	0.0000	51.8649	0.9472	1696.8	13	10416.4	10548.7
10	origin_group{Canada&Botswana Sort-Angola&Recycled}	Entered	0.0000	30.54927	0.9475	1223.1	14	9949.77	10091.6
11	shape{Oval-Radiant}	Entered	0.0000	22.3244	0.9477	877.51	15	9607.87	9759.16
12	shape{Heart-Asscher}	Entered	0.0000	17.23852	0.9478	611.09	16	9343.46	9504.2
13	symmetry 2{Good&Fair-Very Good}	Entered	0.0000	28.08199	0.9481	177.82	18	8911.86	9091.51
14	Polish 2.0{Good or Below-Very Good}	Entered	0.0000	5.526438	0.9481	93.762	19	8827.89	9017
15	origin_group{Angola-Recycled}	Entered	0.0000	3.01578	0.9481	48.803	20	8782.95	8981.51
16	origin_group{Canada-Botswana Sort}	Entered	0.0000	1.592937	0.9481	25.999	21	8760.15	8968.16
17	symmetry 2{Good-Fair}	Entered	0.0143	0.38524	0.9481	22	22	8756.15	8973.62

B.8 — Stepwise Regression: Lab-Grown Diamonds Only

Stepwise Fit for log_price — Student Edition

Stepwise Regression Control

SSE	DFE	RMSE	RSquare	RSquare Adj	Cp	p	AICc	BIC
5017.3202	93113	0.2321297	0.9439	0.9439	17.178918	18	-7713.66	-7534.28

Current Estimates

Lock	Entered	Parameter	Estimate	nDF	SS	"F Ratio"	"Prob>F"
☑	☑	Intercept	6.14460547	1	0	0.000	1
☐	☑	shape{Pear&Round&Marquise-Heart&Princess&Asscher&Emerald&Oval&Radiant&Cushion}	0.00975016	9	72.54238	149.585	4e-282
☐	☑	shape{Pear&Round-Marquise}	0.02133773	2	45.13365	418.802	9e-182
☐	☑	shape{Pear-Round}	0.03723437	1	42.15989	782.417	2e-171
☐	☑	shape{Heart&Princess&Asscher&Emerald&Oval-Radiant&Cushion}	-0.0026388	6	24.46021	75.657	1.2e-94
☐	☑	shape{Heart&Princess-Asscher&Emerald&Oval}	-0.0201694	4	22.88135	106.160	2.1e-90
☐	☑	shape{Heart-Princess}	-0.0207673	1	3.246476	60.249	8.4e-15
☐	☑	shape{Asscher&Emerald-Oval}	-0.0210648	2	9.164659	85.040	1.3e-37
☐	☑	shape{Asscher-Emerald}	-0.0244871	1	4.53327	84.130	4.7e-20
☐	☑	shape{Radiant-Cushion}	0.00404114	1	0.248478	4.611	0.03176
☐	☑	Polish 2.0{Good or Below&Very Good-Excellent}	-0.0225425	2	17.85161	165.648	1.5e-72
☐	☑	Polish 2.0{Good or Below-Very Good}	0.02756883	1	0.353664	6.563	0.01041
☐	☑	symmetry 2{Fair&Good&Very Good-Excellent}	0.03762651	2	19.17384	177.917	7.6e-78
☐	☑	symmetry 2{Fair&Good-Very Good}	0.01187153	1	0.23448	4.352	0.03698
☐	☐	symmetry 2{Fair-Good}	0	1	0.009641	0.179	0.67231
☐	☑	log_carat	1.31144692	1	66405.34	1232371	0
☐	☑	cut_n	-0.0102586	1	2.59167	48.097	4.1e-12
☐	☑	color_n	0.06037025	1	796.5155	14781.98	0
☐	☑	clarity_n	0.10129707	1	1114.462	20682.54	0

Step History

Step	Parameter	Action	"Sig Prob"	Seq SS	RSquare	Cp	p	AICc	BIC
1	log_carat	Entered	0.0000	81644.98	0.9132	50918	2	32889.3	32917.6
2	clarity_n	Entered	0.0000	1870.698	0.9341	16203	3	7208.54	7246.3
3	color_n	Entered	0.0000	756.6164	0.9426	2163.9	4	-5591.5	-5544.3
4	shape{Pear-Round}	Entered	0.0000	61.6085	0.9433	1026.6	7	-6709.8	-6634.2
5	symmetry 2{Fair&Good&Very Good-Excellent}	Entered	0.0000	14.40785	0.9434	761.18	8	-6972.6	-6887.7
6	Polish 2.0{Good or Below&Very Good-Excellent}	Entered	0.0000	14.07522	0.9436	501.97	9	-7230.1	-7135.7
7	shape{Heart&Princess-Asscher&Emerald&Oval}	Entered	0.0000	11.8264	0.9437	286.49	11	-7444.7	-7331.4
8	shape{Asscher&Emerald-Oval}	Entered	0.0000	5.06445	0.9438	194.5	12	-7536.5	-7413.8
9	shape{Asscher-Emerald}	Entered	0.0000	3.786819	0.9438	126.23	13	-7604.7	-7472.5
10	shape{Heart-Princess}	Entered	0.0000	2.536735	0.9438	81.149	14	-7649.7	-7508.1
11	cut_n	Entered	0.0000	2.897414	0.9439	29.378	15	-7701.5	-7550.4
12	Polish 2.0{Good or Below-Very Good}	Entered	0.0024	0.496027	0.9439	22.173	16	-7708.7	-7548.2
13	shape{Radiant-Cushion}	Entered	0.0312	0.250155	0.9439	19.53	17	-7711.3	-7541.4
14	symmetry 2{Fair&Good-Very Good}	Entered	0.0370	0.23448	0.9439	17.179	18	-7713.7	-7534.3

B.9 — Logistic Regression: Predicting Botswana Origin

Nominal Logistic Fit for Botswana_Dummy

Student Edition

Effect Summary

Converged in Gradient, 4 iterations

Iterations

Whole Model Test

Model	-LogLikelihood	DF	ChiSquare	Prob>ChiSq
Difference	7746.743	18	15493.49	<.0001*
Full	57642.673			
Reduced	65389.416			

RSquare (U)	0.1185
AICc	115323
BIC	115503
Observations (or Sum Wgts)	94448

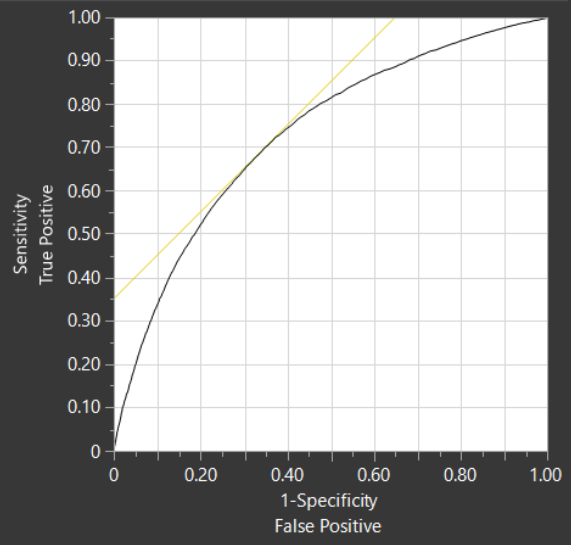
Fit Details

Lack Of Fit

Parameter Estimates

Effect Likelihood Ratio Tests

Receiver Operating Characteristic



Using Botswana_Dummy='1' to be the positive level

AUC
0.73079

Confusion Matrix

Training

Actual	Predicted Count	
Botswana_Dummy	1	0
1	34935	14195
0	16340	28978

Actual	Predicted Rate	
Botswana_Dummy	1	0
1	0.711	0.289
0	0.361	0.639